50Days-coding-challenge

Day-36[Python]

Student Batch Performance Analyzer

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Create user-defined functions in Python to analyze student performance data.

Your functions should:

1. **calculate_average(values)**

Return the average of a list of numbers.

2. **find_top_performers(data, column, threshold)**

Return students whose score in the given column is greater than or equal to the threshold.

3. **find_at_risk_students(data, column, threshold)**

Return students whose score in the given column is below the threshold.

4. **batch_summary(data)**

Return summary insights such as:

- **total students**
- **average attendance**
- **average assignment submission**
- **average total consumption**

5. **rank_batches(batch_data)**

Compare multiple batches and return the best-performing batch.

Sample challenge statement:

You are given student performance data for three batches: b7, b8, and b9.

Write Python user-defined functions to calculate batch-level insights and identify:

- **best batch overall**
- **top-performing students**
- **at-risk students**
- **average attendance**
- **average assignment submission**
- **average content consumption**

Expected output:

Best Batch: b9

Batch Summary:

b7 - Avg Consumption: 46.4%, Avg Attendance: 56.6% b8 - Avg Consumption: 48.3%, Avg Attendance: 62.3% b9 - Avg Consumption: 51.7%, Avg Attendance: 69.1%

At-Risk Students:

Students with consumption below 20%

This is a practical data analyst challenge because it tests: functions, lists, dictionaries, conditions, loops, aggregation, and basic analytics logic.

DataSet(List of Dictionaries)


```
data = [
   {"batch": "b7", "name": "Anil", "consumption": 72, "live": 65, "recording": 70,
"assignment_sub": 80, "assignment_score": 75},
   {"batch": "b7", "name": "Meena", "consumption": 45, "live": 50, "recording": 55,
"assignment_sub": 40, "assignment_score": 35},
   {"batch": "b7", "name": "Ravi", "consumption": 18, "live": 20, "recording": 25,
"assignment_sub": 10, "assignment_score": 15},
   {"batch": "b7", "name": "Pooja", "consumption": 85, "live": 75, "recording": 80,
"assignment_sub": 90, "assignment_score": 88},
   {"batch": "b7", "name": "Kiran", "consumption": 30, "live": 40, "recording": 45,
"assignment_sub": 25, "assignment_score": 28},

   {"batch": "b8", "name": "Asha", "consumption": 90, "live": 85, "recording": 88,
"assignment_sub": 92, "assignment_score": 90},
   {"batch": "b8", "name": "Manoj", "consumption": 55, "live": 60, "recording": 65,
"assignment_sub": 50, "assignment_score": 45},
   {"batch": "b8", "name": "Deepa", "consumption": 22, "live": 30, "recording": 35,
"assignment_sub": 20, "assignment_score": 18},
   {"batch": "b8", "name": "Rahul", "consumption": 78, "live": 70, "recording": 75,
"assignment_sub": 80, "assignment_score": 76},
   {"batch": "b8", "name": "Sneha", "consumption": 10, "live": 15, "recording": 20,
```

"assignment_sub": 5, "assignment_score": 8},

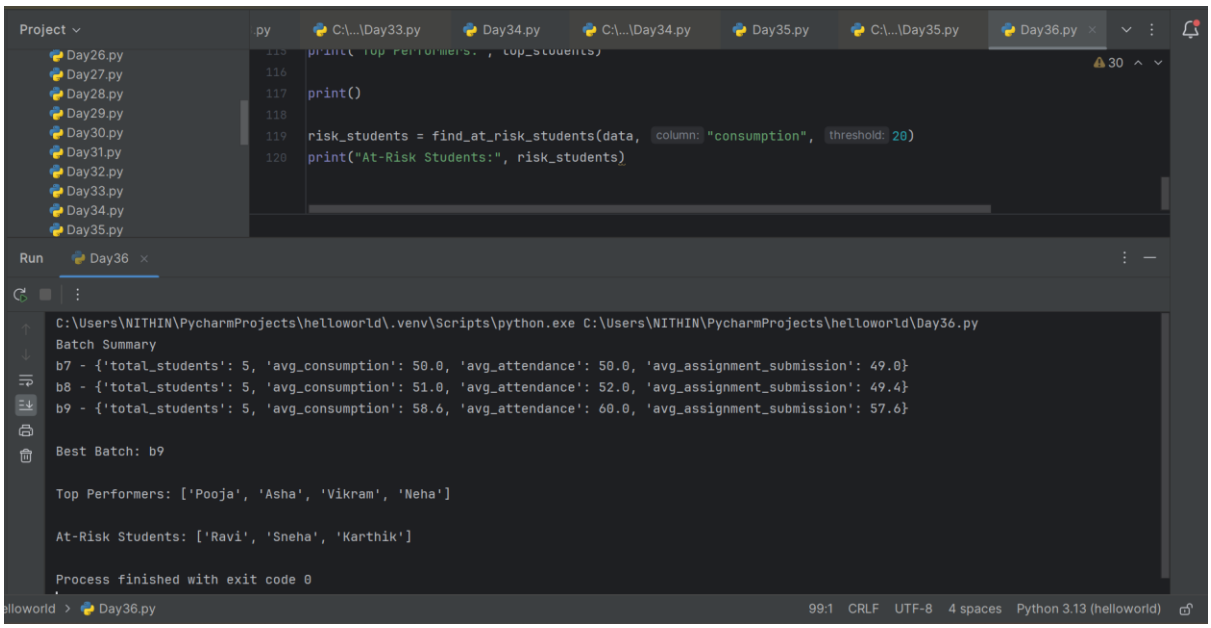
{"batch": "b9", "name": "Vikram", "consumption": 88, "live": 82, "recording": 90, "assignment_sub": 85, "assignment_score": 80},

{"batch": "b9", "name": "Divya", "consumption": 60, "live": 68, "recording": 72, "assignment_sub": 65, "assignment_score": 60},

{"batch": "b9", "name": "Arjun", "consumption": 35, "live": 40, "recording": 50, "assignment_sub": 30, "assignment_score": 25},

{"batch": "b9", "name": "Neha", "consumption": 95, "live": 90, "recording": 95, "assignment_sub": 98, "assignment_score": 96},

{"batch": "b9", "name": "Karthik", "consumption": 15, "live": 20, "recording": 25, "assignment_sub": 10, "assignment_score": 12}

]

Final output: